MA3211S (AY25/26)
Complex Analysis I (S)
by Prof Dinh Tien Cuong
Compiled by ygh3rn

Def. 1 (Topological Spaces)
A *topological space* is a set X together with a collection τ of subsets of X satisfying:
- $\varnothing, X \in \tau$;
- If $U_1, \dots, U_n \in \tau$, then $U_1 \cap \cdots \cap U_n \in \tau$;
- Any union of sets in τ is in τ.

Def. 2 (Topology)
τ is called a *topology* on X. Sets in τ are called *open*. A subset of X is *closed* if its complement is open.

Def. 3 (Closure)
Let $A \subset X$. The *closure* of A, denoted $\overline{A}$, is the intersection of all closed sets containing A.

Def. 4 (Interior)
Let $A \subset X$. The *interior* of A, denoted $\operatorname{int} A$ or A°, is the union of all open sets contained in A.

Def. 5 (Boundary)
The *boundary* of a set A is $\partial A = \overline{A} \setminus \operatorname{int} A$.

Def. 6 (Convergence)
Let (x_n) be a sequence in a topological space (X, τ). We say $x_n \to x$ if for every open set U with $x \in U$, there exists N such that $n > N \Rightarrow x_n \in U$.

Def. 7 (Continuity)
Let (X, τ) and (Y, σ) be topological spaces and $f : X \to Y$. f is *continuous* if for every $U \in \sigma$, $f^{-1}(U) \in \tau$.

Def. 8 (Neighbourhood)
Let (X, τ) be a topological space and $x \in X$. A *neighbourhood* of x is a set containing an open set that contains x.

Prop. 9 (Continuity via Neighbourhoods)
Let (X, τ) and (Y, σ) be topological spaces and $f : X \to Y$. The following are equivalent:
1. f is continuous.
2. For every $x \in X$ and every neighbourhood M of $f(x)$, there exists a neighbourhood N of x such that $f(N) \subset M$.

Def. 10 (Homeomorphism)
Let (X, τ) and (Y, σ) be topological spaces and $f : X \to Y$. f is a *homeomorphism* if it is continuous and has a continuous inverse.

Def. 11 (Subspace Topology)
Let (X, τ) be a topological space and $Y \subset X$. The *subspace topology* on Y is $\{U \cap Y : U \in \tau\}$.

Prop. 12 (Open Sets in Subspaces)
Let (X, d) be a metric space, $Y \subset X$, and $U \subset Y$. The following are equivalent:
1. U is open in the subspace topology on Y.
2. For every $u \in U$, there exists $\delta > 0$ such that if $v \in Y$ and $d(u, v) < \delta$, then $v \in U$.

Def. 13 (Quotient Topology)
Let (X, τ) be a topological space and $\sim$ an equivalence relation on X. Let $Y = X/\sim$ and let $q : X \to Y$ be the quotient map sending x to its equivalence class. The *quotient topology* on Y is $\{U \subset Y : q^{-1}(U) \in \tau\}$.

Def. 14 (Product Topology)
Let (X, τ) and (Y, σ) be topological spaces. The *product topology* on $X \times Y$ is the collection of all unions of sets of the form $U \times V$ with $U \in \tau$ and $V \in \sigma$.

Def. 15 (Product Topology on Finite Products)
Let $(X_i, \tau_i)_{i=1}^n$ be topological spaces. The product topology on $\prod_{i=1}^n X_i$ is the collection of all unions of sets of the form $\prod_{i=1}^n U_i$ with $U_i \in \tau_i$.

Def. 16 (Basis)
Let (X, τ) be a topological space. A *basis* for τ is a subset $\mathcal{B} \subset \tau$ such that every $U \in \tau$ can be written as a union of sets in $\mathcal{B}$. Elements of $\mathcal{B}$ are called *basic open sets*.

Def. 17 (Basis of Neighbourhoods)
Let (X, τ) be a topological space and $x \in X$. A *basis of neighbourhoods* of x is a collection $\mathcal{N}$ of neighbourhoods of x such that every neighbourhood of x contains some $N \in \mathcal{N}$.

Prop. 18 (Quotient and Product Topologies)
The quotient topology and the product topology are topologies.

Def. 19 (Open Cover)
Let (X, τ) be a topological space. An *open cover* of X is a collection $\{U_\gamma : \gamma \in \Gamma\}$ of open sets such that $X = \bigcup_{\gamma \in \Gamma} U_\gamma$. If $Y \subset X$, an open cover of Y satisfies $Y \subset \bigcup_{\gamma \in \Gamma} U_\gamma$.

Def. 20 (Subcover)
A *subcover* of an open cover $\mathcal{U}$ is a subcollection of $\mathcal{U}$ that still covers the set.

Def. 21 (Compactness)
A topological space is *compact* if every open cover has a finite subcover.

Lem. 22 (Compactness in Subspaces)
Let (X, τ) be a topological space and $Y \subset X$. The following are equivalent:
1. Y is compact in the subspace topology.
2. Every cover of Y by open sets in τ has a finite subcover.

Thm. 23 (Heine–Borel Theorem)
The closed interval $[a, b] \subset \mathbb{R}$ is compact.

Thm. 24 (Continuous Image of Compact Sets)
Let X be compact and $f : X \to Y$ continuous. Then $f(X)$ is compact.

Thm. 25 (Closed Subsets of Compact Sets)
Let X be compact and $K \subset X$ closed. Then K is compact.

Def. 26 (Hausdorff Space)
A topological space (X, τ) is *Hausdorff* if for any distinct $x, y \in X$, there exist disjoint open sets U, V such that $x \in U$ and $y \in V$.

Thm. 27 (Compact Sets in Hausdorff Spaces)
Every compact subset of a Hausdorff topological space is closed.

Thm. 28 (Finite Products of Compact Sets)
A product of finitely many compact topological spaces is compact.

Thm. 29 (Heine–Borel Theorem in $\mathbb{R}^n$)
A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Thm. 30 (Extreme Value Theorem)
A continuous real-valued function on a compact metric space is bounded and attains its maximum and minimum values.

Thm. 31 (Uniform Continuity)
Let X be a compact metric space, Y a metric space, and $f : X \to Y$ continuous. Then f is uniformly continuous.

Thm. 32 (Distance to Closed Sets)
Let X be a metric space, $K \subset X$ compact, and $Y \subset X$ closed. Then there exists $x \in K$ such that

$$d(x, Y) = \inf\{d(w, y) : w \in K,\ y \in Y\}.$$

In particular, if $K \cap Y = \varnothing$, then $d(K, Y) > 0$.

Def. 33 (Sequential Compactness)
A topological space X is *sequentially compact* if every sequence in X has a convergent subsequence converging to a point in X.

Thm. 34 (Compact Metric Spaces)
Every compact metric space is sequentially compact.

Thm. 35 (Compactness in $\mathbb{R}^n$)
Let $X \subset \mathbb{R}^n$. The following are equivalent:
1. X is compact.
2. X is sequentially compact.
3. X is closed and bounded.

Def. 36 (Disconnected Sets and Connected Spaces)
Let X be a topological space. Suppose $U, V \subset X$ satisfy:
1. U and V are open,
2. $U \cap V = \varnothing$,
3. $X = U \cup V$,
4. $U \neq \varnothing$ and $V \neq \varnothing$.

Then U and V are said to *disconnect* X. The space X is *connected* if no such pair exists.

Def. 37 (Disconnected Subspaces)
Let Y be a subspace of a topological space X. Then Y is disconnected in the subspace topology if and only if there exist open sets $U, V \subset X$ such that

$$U \cap Y \neq \varnothing, \quad V \cap Y \neq \varnothing, \quad Y \subset U \cup V, \quad (U \cap V \cap Y) = \varnothing.$$

In this case U and V are said to disconnect Y.

Prop. 38 (Characterisations of Connectedness)
Let X be a topological space. The following are equivalent:
1. X is connected.
2. Every continuous function $f : X \to \mathbb{Z}$ is constant.
3. The only subsets of X that are both open and closed are $\varnothing$ and X.

Prop. 39
A continuous image of a connected space is connected.

Def. 40 (Intervals in $\mathbb{R}$)
A subset $I \subset \mathbb{R}$ is an *interval* if whenever $x \le y \le z$ and $x, z \in I$, then $y \in I$.

Thm. 41 (Connected Sets in $\mathbb{R}$)
A subset of $\mathbb{R}$ is connected if and only if it is an interval.

Thm. 42 (Intermediate Value Theorem)
Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous. If $f(a) < y < f(b)$, then there exists $x \in [a, b]$ such that $f(x) = y$.

Def. 43 (Path and Path-Connectedness)
Let X be a topological space and $x, y \in X$. A *path* from x to y is a continuous map $\phi : [a, b] \to X$ with $\phi(a) = x$ and $\phi(b) = y$. The space X is *path-connected* if such a path exists for all $x, y \in X$.

Prop. 44
Every path-connected topological space is connected.

Def. 45 (Join and Reverse of Paths)
Let $\phi : [a, b] \to X$ and $\psi : [c, d] \to X$ be paths with $\phi(b) = \psi(c)$. The *join* $\phi \vee \psi : [a, b + d - c] \to X$ is defined by

$$(\phi \vee \psi)(t) = \begin{cases} \phi(t), & a \le t < b, \\ \psi(t + c - b), & b \le t \le b + d - c. \end{cases}$$

The *reverse* of ϕ is the path $-\phi : [-b, -a] \to X$ given by $(-\phi)(t) = \phi(-t)$.

Def. 46 (Path Relation and Path Components)
Let X be a topological space. Write $x \to y$ if there exists a continuous path $\phi : [a, b] \to X$ with $\phi(a) = x$ and $\phi(b) = y$. This defines an equivalence relation on X. The equivalence classes are called the *path-components* of X.

Def. 47 (Polygonal Path)
Let $X \subset \mathbb{R}^n$. A *polygonal path* in X is a continuous path $\phi : [a, b] \to X$ that is piecewise linear, i.e. there exist

$$a = x_0 < x_1 < \cdots < x_m = b$$

such that for each $i = 1, \dots, m$ and $t \in [0, 1]$,

$$\phi((1 - t)x_{i-1} + t x_i) = (1 - t)\phi(x_{i-1}) + t\phi(x_i).$$

The set X is *polygonally connected* if any two points of X can be joined by a polygonal path. The relation $x \rightsquigarrow y$ defined by the existence of such a path is an equivalence relation; its equivalence classes are the *polygonal components* of X.

Thm. 48 (Connectedness in $\mathbb{R}^n$)
Let $X \subset \mathbb{R}^n$ be open. The following are equivalent:
1. X is connected;
2. X is path-connected;
3. X is polygonally connected.

Thm. 49 (Path Components of Open Sets)
If $X \subset \mathbb{R}^n$ is open, then all its path-components are open.

Lem. 50 (Complement of Compact Sets in $\mathbb{R}^n$)
Let $n \ge 2$ and let $X \subset \mathbb{R}^n$ be compact. Then X^c has only open path-components, and exactly one of them is unbounded.

Def. 51 (C^1 Path)
A path $\phi : [a, b] \to \mathbb{C}$ is called C^1 if ϕ is continuously differentiable on (a, b) and has appropriate one-sided derivatives at a and b. In particular, ϕ and ϕ' are bounded.

Def. 52 (Equivalent C^1 Paths)
Two C^1 paths $\phi : [a, b] \to \mathbb{C}$ and $\psi : [c, d] \to \mathbb{C}$ are *equivalent* if there exists a C^1 function $\gamma : [a, b] \to [c, d]$ such that

$$\gamma'(t) > 0 \ \forall t \in [a, b], \qquad \gamma(a) = c,\ \gamma(b) = d,$$

and

$$\phi(t) = \psi(\gamma(t)) \quad \forall t \in [a, b].$$

Def. 53 (Track of a Path)
For a path $\phi : [a, b] \to \mathbb{C}$, its *track* is

$$\phi^* = \{\phi(t) : a \le t \le b\}.$$

Equivalent paths have the same track.

Def. 54 (Piecewise C^1 Path)
A path $\phi : [a, b] \to \mathbb{C}$ is *piecewise* C^1 if there exist $a = x_0 < x_1 < \cdots < x_n = b$ such that each restriction $\phi|_{[x_{i-1}, x_i]}$ is C^1. Equivalently, ϕ is the join of finitely many C^1 paths.

Def. 55 (Equivalent Piecewise C^1 Paths)
Two piecewise C^1 paths $\phi : [a, b] \to \mathbb{C}$ and $\psi : [c, d] \to \mathbb{C}$ are *equivalent* if they admit partitions $a = x_0 < \cdots < x_n = b$ and $c = y_0 < \cdots < y_n = d$ such that for each i,

$$\phi|_{[x_{i-1}, x_i]} \sim \psi|_{[y_{i-1}, y_i]}$$

as C^1 paths.

Def. 56 (Closed and Simple Paths)
A piecewise C^1 path $\phi : [a, b] \to \mathbb{C}$ is *closed* if $\phi(a) = \phi(b)$. It is *simple* if $\phi(x) = \phi(y)$ implies $x = y$ or $\{x, y\} = \{a, b\}$.

Def. 57 (Complex Riemann Integral)
A function $f : [a, b] \to \mathbb{C}$ is Riemann integrable if $\Re f$ and $\Im f$ are Riemann integrable, and

$$\int_a^b f(t)\, dt = \int_a^b \Re f(t)\, dt + i \int_a^b \Im f(t)\, dt.$$

Lem. 58 (Triangle Inequality for Integrals)
If $f : [a, b] \to \mathbb{C}$ is continuous, then

$$\left| \int_a^b f(t)\, dt \right| \le \int_a^b |f(t)|\, dt.$$

Def. 59 (Domain)
A *domain* is a connected open subset of $\mathbb{C}$.

Def. 60 (Line Segment Path)
For $x, y \in \mathbb{C}$, let $[x \to y]$ denote the path $\phi : [0, 1] \to \mathbb{C}$ defined by $\phi(t) = (1 - t)x + ty$.

Def. 61 (Convex Domain)
A domain D is *convex* if for all $x, y \in D$,

$$[x \to y]^* \subset D.$$

Def. 62 (Star-Shaped Domain)
A domain D is *star-shaped* if there exists $z_0 \in D$ such that

$$[z_0 \to z]^* \subset D \quad \forall z \in D.$$

Def. 63 (**Complex Path Integral**)
Let D be a domain, $f : D \to \mathbb{C}$ continuous, and $\phi : [a, b] \to D$ a C^1 path. The *integral of f along ϕ* is

$$\int_\phi f(z)\,dz = \int_a^b f(\phi(t))\,\phi'(t)\,dt.$$

If ϕ is piecewise C^1, the integral is defined as the sum over its pieces.

Lem. 64 (**Invariance under Reparametrisation**)
Equivalent (piecewise) C^1 paths have the same complex line integral.

Def. 65 (**Length of a C^1 Path**)
Let D be a domain and $\phi : [a, b] \to D$ be a C^1 path. The length of ϕ is

$$L(\phi) = \int_a^b |\phi'(t)|\,dt.$$

Lem. 66 (**Estimation Lemma**)
Let D be a domain, $f : D \to \mathbb{C}$ be continuous, and $\phi : [a, b] \to D$ be a C^1 path. Then

$$\left|\int_\phi f(z)\,dz\right| \le \sup_{z\in\phi^*} |f(z)|\,L(\phi).$$

Rem. 67
The estimation lemma extends to piecewise C^1 paths. Henceforth, all paths are assumed piecewise C^1.

Prop. 68 (**Fundamental Theorem of Calculus**)
Let D be a domain and $f : D \to \mathbb{C}$ be continuous. If f has an antiderivative F on D and $\phi : [a, b] \to D$ is a path, then

$$\int_\phi f(z)\,dz = F(\phi(b)) - F(\phi(a)).$$

Cor. 69
If D is a domain, $f : D \to \mathbb{C}$ has an antiderivative, and ϕ is a closed path in D, then

$$\int_\phi f(z)\,dz = 0.$$

Lem. 70 (**Characterisation of Antiderivatives**)
Let D be a star-domain and $f : D \to \mathbb{C}$ be continuous. The following are equivalent:
(1) f has an antiderivative on D.
(2) $\int_\phi f(z)\,dz = 0$ for all closed paths ϕ in D.
(3) $\int_{\partial T} f(z)\,dz = 0$ for every triangle $T \subset D$.

Def. 71 (**Holomorphic Function**)
Let D be a domain. A function $f : D \to \mathbb{C}$ is called holomorphic (or $\mathbb{C}$-analytic) if it is complex differentiable at every point of D.

Thm. 72 (**Cauchy's Theorem for Triangles**)
Let D be a domain and $T \subset D$ be a triangle. If $f : D \to \mathbb{C}$ is $\mathbb{C}$-analytic, then

$$\int_{\partial T} f(z)\,dz = 0.$$

Cor. 73 (**Cauchy's Theorem for a Star-Domain**)
Let D be a star-domain and $f : D \to \mathbb{C}$ be $\mathbb{C}$-analytic. Then

$$\int_\phi f(z)\,dz = 0$$

for all closed paths ϕ in D.

Def. 74 (**Homotopy**)
Let $\phi, \psi : [0, 1] \to D$ be piecewise C^1 closed paths in a domain D. A homotopy from ϕ to ψ is a continuous function

$$\gamma : [0, 1]^2 \to D$$

such that:
- $\gamma(0, t) = \phi(t)$,
- $\gamma(1, t) = \psi(t)$,
- for each $s \in [0, 1]$, the path $\gamma_s(t) = \gamma(s, t)$ is closed and piecewise C^1.

Def. 75 (**Elementary Deformation**)
A closed path ψ is an elementary deformation of a closed path ϕ if there exist $0 = x_0 < x_1 < \cdots < x_n = 1$ and convex open sets $C_1, \ldots, C_n \subset D$ such that

$$x_{i-1} \le t \le x_i \;\Rightarrow\; \phi(t), \psi(t) \in C_i.$$

Lem. 76 (**Invariance under Elementary Deformation**)
Let D be a domain and $f : D \to \mathbb{C}$ be $\mathbb{C}$-analytic. If ψ is an elementary deformation of a closed path ϕ, then

$$\int_\phi f(z)\,dz = \int_\psi f(z)\,dz.$$

Prop. 77 (**Homotopy via Elementary Deformations**)
Let $\phi, \psi : [0, 1] \to D$ be homotopic closed paths in a domain D. Then there exist closed paths

$$\phi = \phi_0, \phi_1, \ldots, \phi_n = \psi$$

such that ϕ_i is an elementary deformation of ϕ_{i-1}.

Cor. 78 (**Homotopy Invariance of Integrals**)
Let D be a domain and $f : D \to \mathbb{C}$ be $\mathbb{C}$-analytic. If ϕ and ψ are homotopic closed paths in D, then

$$\int_\phi f(z)\,dz = \int_\psi f(z)\,dz.$$

Def. 79 (**Contractible Path**)
A closed path ϕ in a domain D is contractible if it is homotopic to a constant path.

Def. 80 (**Simply Connected Domain**)
A domain D is simply connected if every closed path in D is contractible.

Cor. 81 (**Cauchy's Theorem for Simply Connected Domains**)
Let D be a domain and $f : D \to \mathbb{C}$ be $\mathbb{C}$-analytic.
- If a closed path ϕ is contractible, then
$$\int_\phi f(z)\,dz = 0.$$
- If D is simply connected, then
$$\int_\phi f(z)\,dz = 0$$
for all closed paths ϕ in D.

Def. 82 (**Discs and Circles**)
For $z_0 \in \mathbb{C}$ and $r > 0$, define

$$B_r(z_0) = \{z \in \mathbb{C} : |z - z_0| < r\},$$

$$C_r(z_0) : t \mapsto z_0 + re^{2\pi i t},\ t \in [0, 1].$$

Thm. 83 (**Cauchy's Integral Formula**)
Let D be a domain and $f : D \to \mathbb{C}$ be $\mathbb{C}$-analytic. Assume $\overline{B_r(z_0)} \subset D$. Then for any $z \in \mathbb{C}$ with $z \ne w$ for all $w \in C_r(z_0)$,

$$\frac{1}{2\pi i}\int_{C_r(z_0)} \frac{f(w)}{w - z}\,dw = \begin{cases} f(z), & z \in B_r(z_0),\\ 0, & z \notin \overline{B_r(z_0)}.\end{cases}$$

Thm. 84 (**Circle Integral Identity**)
Let $z_0 \in \mathbb{C}$. Then for any integer $m \in \mathbb{Z}$,

$$\frac{1}{2\pi i}\int_{C_r(z_0)} (w - z_0)^m\,dw = \begin{cases} 1, & m = -1,\\ 0, & m \ne -1.\end{cases}$$

Thm. 85 (**Liouville's Theorem**)
Every bounded entire (analytic on the whole $\mathbb{C}$) function is constant.

Thm. 86 (**Fundamental Theorem of Algebra**)
Every non-constant polynomial has at least one root in $\mathbb{C}$.

Prop. 87 (**Cauchy-Type Integral Defines Holomorphic Function**)
Let g be continuous on $\{z \in \mathbb{C} : |z - z_0| = r\}$ and define, for $z \in B_r(z_0)$,

$$f(z) = \int_{C_r(z_0)} \frac{g(w)}{(w - z)^n}\,dw.$$

Then f is $\mathbb{C}$-analytic on $B_r(z_0)$ and

$$f'(z) = n\int_{C_r(z_0)} \frac{g(w)}{(w - z)^{n+1}}\,dw.$$

Cor. 88 (**Cauchy Differentiation Formula**)
Let D be a domain and $f : D \to \mathbb{C}$ $\mathbb{C}$-analytic. If $B_r(z_0) \subset D$, then f is infinitely differentiable on $B_r(z_0)$ and for all $z \in B_r(z_0)$,

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_{C_r(z_0)} \frac{f(w)}{(w - z)^{n+1}}\,dw.$$

Cor. 89 (**Cauchy Estimates**)
Let $M_r = \max_{|w-z_0|=r} |f(w)|$. Then for every $n \ge 0$,

$$|f^{(n)}(z_0)| \le \frac{n!}{r^n}\,M_r.$$

Thm. 90 (**Morera's Theorem**)
Let D be a domain and $f : D \to \mathbb{C}$ continuous. If

$$\int_{\partial T} f(z)\,dz = 0$$

for every triangle $T \subset D$, then f is $\mathbb{C}$-analytic on D.

Lem. 91 (**Uniform Convergence of Power Series on Disks**)
Consider a power series $\sum_{n=0}^\infty a_n(z - z_0)^n$. If it converges at some point with $|z - z_0| = \rho$, then it converges for all $|w - z_0| < \rho$, and for every $r < \rho$ the convergence is uniform on $B_r(z_0)$.

Def. 92 (**Radius of Convergence**)
The radius of convergence of $\sum_{n=0}^\infty a_n(z - z_0)^n$ is

$$R = \sup\Big\{r : \exists z \text{ with } |z - z_0| = r \text{ and } \sum_{n=0}^\infty a_n(z - z_0)^n \text{ converges}\Big\}.$$

Lem. 93 (**Uniform Limit and Path Integrals**)
Let D be a domain, $\phi : [a, b] \to D$ a path, and $f_n : D \to \mathbb{C}$ continuous. If $f_n \to f$ uniformly on ϕ^*, then

$$\int_\phi f_n(z)\,dz \to \int_\phi f(z)\,dz.$$

Lem. 94 (**Uniqueness of Power Series Coefficients**)
Let $f(z) = \sum_{n=0}^\infty a_n(z-z_0)^n$ and $g(z) = \sum_{n=0}^\infty b_n(z-z_0)^n$. If there exists a sequence $z_k \to z_0$ with $z_k \ne z_0$ and $f(z_k) = g(z_k)$ for all k, then $a_n = b_n$ for all n.

Lem. 95 (**Holomorphicity of Power Series**)
Let $f(z) = \sum_{n=0}^\infty a_n(z - z_0)^n$ with radius of convergence R. Then f is $\mathbb{C}$-analytic on $B_R(z_0)$ and

$$f'(z) = \sum_{n=1}^\infty n a_n(z - z_0)^{n-1}.$$

Thm. 96 (**Taylor's Theorem**)
Let D be a domain and $f : D \to \mathbb{C}$ holomorphic. If $B_R(z_0) \subset D$, then there exist unique coefficients $(a_n)_{n\ge0}$ such that

$$f(z) = \sum_{n=0}^\infty a_n(z - z_0)^n \qquad \forall z \in B_R(z_0).$$

The coefficients are given by

$$a_n = \frac{1}{2\pi i}\int_{C_\rho(z_0)} \frac{f(w)}{(w - z_0)^{n+1}}\,dw, \qquad 0 < \rho < R,$$

and equivalently

$$a_n = \frac{f^{(n)}(z_0)}{n!}.$$

Thm. 97 (**Identity Theorem**)
Let D be a domain and $f, g : D \to \mathbb{C}$ $\mathbb{C}$-analytic. If $z_k \to z_0 \in D$ with $z_k \ne z_0$ and $f(z_k) = g(z_k)$ for all k, then $f \equiv g$ on D. In particular, the zeros of a non-constant $\mathbb{C}$-analytic function are isolated.

Prop. 98 (**Zeros Have Finite Order**)
Let D be a domain, $z_0 \in D$, and $f : D \to \mathbb{C}$ $\mathbb{C}$-analytic with $f \not\equiv 0$. Then there exist unique $k \in \mathbb{N}$ and a unique $\mathbb{C}$-analytic g with $g(z_0) \ne 0$ such that

$$f(z) = (z - z_0)^k g(z).$$

Thm. 99 (**Riemann's Removable Singularity Theorem**)
Let D be a domain, $z_0 \in D$, and let $f : D\setminus\{z_0\} \to \mathbb{C}$ be analytic. If f is bounded near z_0, then $\lim_{z\to z_0} f(z) = a$ exists and the function

$$g(z) = \begin{cases} f(z), & z \ne z_0,\\ a, & z = z_0,\end{cases}$$

is analytic on D.

Prop. 100 (**Pole Characterisation**)
Let D be a domain, $z_0 \in D$, and $f : D \setminus \{z_0\} \to \mathbb{C}$ analytic. If $|f(z)| \to \infty$ as $z \to z_0$, then there exists a unique $k \ge 1$ and a unique analytic $g : D \to \mathbb{C}$ with $g(z_0) \ne 0$ such that

$$f(z) = (z - z_0)^{-k} g(z).$$

Def. 101 (**Classification of Isolated Singularities**)
Let D be a domain, $z_0 \in D$, and $f : D \setminus \{z_0\} \to \mathbb{C}$ analytic.
- z_0 is removable if f is bounded near z_0.
- z_0 is a pole of order k if $f(z) = (z - z_0)^{-k} g(z)$ with $g(z_0) \ne 0$.
- Otherwise z_0 is essential.

Thm. 102 (**Classification at Infinity**)
Let f be entire, and let $g(z) = f(1/z)$. Then the isolated singularity of g at 0 determines the behaviour of f at ∞:
- if ∞ is a removable singularity of f, then f is bounded near ∞, hence bounded on $\mathbb{C}$, so f is constant;
- if ∞ is a pole of f, then f has polynomial growth at ∞, and therefore f is a polynomial.

Thm. 103 (**Casorati–Weierstrass Theorem**)
Let z_0 be an essential singularity of f. Then for every neighbourhood U of z_0, the set $f(U \setminus \{z_0\})$ is dense in $\mathbb{C}$. Equivalently, for every $w \in \mathbb{C}$, there exists $z_n \to z_0$ such that $f(z_n) \to w$.

Thm. 104 (**Great Picard Theorem**)
Let z_0 be an isolated essential singularity of a holomorphic function f. Then on every punctured neighbourhood of z_0, the function f takes every complex value, with at most one exception, infinitely often.

Thm. 105 (**Laurent's Theorem**)
Let $D = \{z : a < |z - z_0| < b\}$ and let $f : D \to \mathbb{C}$ analytic. Then there exist unique $(a_n)_{n\in\mathbb{Z}}$ such that

$$f(z) = \sum_{n=-\infty}^\infty a_n(z - z_0)^n.$$

Moreover,

$$a_n = \frac{1}{2\pi i}\int_{C_\rho(z_0)} \frac{f(w)}{(w - z_0)^{n+1}}\,dw.$$

Thm. 106 (**Maximum Modulus Principle**)
Let D be a domain and $f : D \to \mathbb{C}$ analytic. If $|f|$ has a local maximum in D, then f is constant.

Def. 107 (**Principal Value**)
For $z \neq 0$, $\operatorname{Log} z = \log|z| + i\theta, \quad -\pi < \theta \leq \pi$. The principal argument is $\operatorname{Arg} z = \theta$ with $-\pi < \theta \leq \pi$.

Def. 108 (**Continuous Branch of** $\log$)
Let D be a domain not containing 0. A continuous branch of log on D is a continuous $f : D \to \mathbb{C}$ such that $e^{f(z)} = z$ for all $z \in D$.

Lem. 109 (**Existence of Antiderivative**)
Let D be simply connected and $f : D \to \mathbb{C}$ analytic. Then f has an antiderivative on D.

Lem. 110 (**Uniqueness of Branch**)
Let D be simply connected, $0 \notin D$. For any $z_0 \in D$ and chosen value of $\log z_0$, there exists a unique continuous branch of log on D taking that value at z_0.

Def. 111 (**Winding Number**)
Let ϕ be a closed path and $z_0 \notin \phi^*$. The winding number of ϕ about z_0 is

$$w(\phi, z_0) = \frac{1}{2\pi i}\int_\phi \frac{dz}{z - z_0}.$$

Prop. 112 (**Properties of Winding Number**)
If $z_0 \notin \phi^*$ then:
- $w(\phi, z_0) \in \mathbb{Z}$,
- $w(\phi, z_0)$ is constant on components of $\mathbb{C} \setminus \phi^*$,
- $w(\phi, z_0) = 0$ on the unbounded component.

Def. 113 (**Residue**)
Let f be $\mathbb{C}$-analytic on a domain D except at finitely many points $z_1, \ldots, z_k$. If near z, $f(w) = \sum_{n=-\infty}^{\infty} a_n(w - z)^n$, then

$$\operatorname{Res}(f, z) = a_{-1}.$$

If $z \notin \{z_1, \ldots, z_k\}$, then $\operatorname{Res}(f, z) = 0$.
Equivalently,

$$\operatorname{Res}(f, z_i) = \frac{1}{2\pi i}\int_{C_\delta(z_i)} f(z)\, dz.$$

Thm. 114 (**Residue at a Pole of Order** m)
Let f have a pole of order m at a. Then

$$\operatorname{Res}(f, a) = \frac{1}{(m-1)!}\lim_{z\to a}\frac{d^{m-1}}{dz^{m-1}}\Big[(z-a)^m f(z)\Big].$$

Thm. 115 (**Residue of** g/h **at a Simple Zero of** h)
Let g and h be holomorphic near a. If

$$f(z) = \frac{g(z)}{h(z)}, \qquad h(a) = 0, \quad h'(a) \neq 0,$$

then a is a simple pole of f and

$$\operatorname{Res}(f, a) = \frac{g(a)}{h'(a)}.$$

Def. 116 (**Principal Part**)
If near z_i, $f(z) = \sum_{n=-\infty}^{\infty} a_n^{(i)}(z - z_i)^n$, the principal part at z_i is

$$g_i(z) = \sum_{n=-\infty}^{-1} a_n^{(i)}(z - z_i)^n.$$

Then $f - g_i$ has a removable singularity at z_i.

Thm. 117 (**Cauchy's Residue Theorem**)
Let D be simply connected and f analytic on D except at finitely many $z_1, \ldots, z_k$. If ϕ is a closed path in D with $\phi^* \cap \{z_1, \ldots, z_k\} = \varnothing$, then

$$\int_\phi f(z)\, dz = 2\pi i \sum_{j=1}^{k} \operatorname{Res}(f, z_j)\, w(\phi, z_j).$$

Def. 118 (**Order**)
If near z_0,

$$f(z) = (z - z_0)^k g(z), \quad g(z_0) \neq 0,$$

then $k = \operatorname{ord}(f, z_0)$.

Thm. 119 (**Principle of the Argument**)
Under the same hypotheses, define

$$ZP(f, \phi) := \frac{1}{2\pi i}\int_\phi \frac{f'(z)}{f(z)}\, dz.$$

Then,

$$ZP(f, \phi) = \sum_{j=1}^{k} \operatorname{ord}(f, z_j)\, w(\phi, z_j).$$

Equivalently,

$$ZP(f, \phi) = \frac{1}{2\pi i}\int_{f\circ\phi} \frac{dz}{z} = w(f \circ \phi, 0).$$

Thm. 120 (**Rouché's Theorem**)
Let f, g be analytic on D except finitely many poles, none on ϕ^*. Assume f and $f + g$ have finitely many zeros, none on ϕ^*, and

$$|g(z)| < |f(z)| \quad \forall z \in \phi^*.$$

Then

$$ZP(f + g, \phi) = ZP(f, \phi).$$

Thm. 121 (**Local Mapping Theorem**)
If f is analytic and non-constant, and $k = \operatorname{ord}(f - w_0, z_0)$, then for small $\varepsilon > 0$ there exists $\delta > 0$ such that for $0 < |w - w_0| < \delta$, there are exactly k solutions of $f(z) = w$ in $0 < |z - z_0| < \varepsilon$.
Equivalently, near z_0 we can write

$$f(z) - w_0 = (z - z_0)^k g(z),$$

where g is analytic and $g(z_0) \neq 0$. Hence locally

$$f(z) - w_0 \approx c(z - z_0)^k$$

for some $c \neq 0$, so f behaves like the map

$$z \mapsto (z - z_0)^k$$

near z_0.

Cor. 122 (**Open Mapping Theorem**)
If f is analytic and non-constant on D, then $f(U)$ is open for every open $U \subset D$.

Def. 123 (**Uniform Convergence of Series**)
Let (f_n) be functions on X. The series $\sum_{n\geq1} f_n$ converges uniformly on X if the partial sums $\sum_{1\leq n\leq N} f_n$ converge uniformly on X as $N \to \infty$.

Def. 124 (**Normal Convergence**)
The series $\sum_{n\geq1} f_n$ converges normally on X if $\sum_{n\geq1}\sup_X |f_n|$ converges. Normal convergence implies uniform convergence.

Def. 125 (**Local Convergence**)
A series converges locally uniformly (resp. locally normally) on a domain Ω if and only if it converges uniformly (resp. normally) on every compact subset of Ω.

Thm. 126 (**Holomorphic Limit of Series**)
Let f_n be holomorphic on a domain Ω. If $\sum_{n\geq1} f_n$ converges uniformly (resp. normally) on all compact subsets of Ω, then $f = \sum_{n\geq1} f_n$ is holomorphic on Ω, and $\sum_{n\geq1} f_n'$ converges uniformly (resp. normally) on compact subsets to f'.

Def. 127 (**Dirichlet Series**)
Let (a_n) be a sequence of complex numbers. Define

$$L(a, s) = \sum_{n\geq1} a_n n^{-s}, \qquad s \in \mathbb{C}.$$

Define

$$\tau = \limsup_{n\to\infty}\frac{\log|a_n|}{\log n}.$$

Prop. 128 (**Dirichlet Series Convergence**)
For every $\delta > 0$, the series $L(a, s)$ converges normally on $\{\Re s > \tau + 1 + \delta\}$ and defines a holomorphic function on $\{\Re s > \tau + 1\}$.

Rem. 129 (**Riemann Zeta Function**)

$$\zeta(s) = \sum_{n\geq1} n^{-s}$$

which is holomorphic for $\Re s > 1$.

Def. 130 (**Meromorphic Function**)
A function f on a domain Ω is meromorphic if it is holomorphic outside a locally finite set $Z \subset \Omega$ and every point of Z is either a removable singularity or a pole.

If g, h are holomorphic and $h \not\equiv 0$, then g/h is meromorphic. Any meromorphic function on Ω is the quotient of two holomorphic functions on Ω.

Def. 131 (**Series of Meromorphic Functions**)
Let f_n be meromorphic on Ω. The series $\sum f_n$ converges uniformly (resp. normally) on compact subsets if for every compact K there exists n_K such that
- f_n has no pole on K for $n \geq n_K$,
- $\sum_{n\geq n_K} f_n$ converges uniformly (resp. normally) on K.

In this case the sum is meromorphic and $\sum f_n'$ converges to f' on compact subsets.

Def. 132 (**Infinite Product**)
Let f_n be holomorphic or meromorphic on Ω. The product $\prod_{n\geq1} f_n$ converges normally on compact subsets if $\sum_{n\geq1}(f_n - 1)$ converges normally on compact subsets.

Thm. 133 (**Infinite Product Theorem**)
If $\prod_{n\geq1} f_n$ converges normally on compact subsets of Ω, then the partial products $\prod_{1\leq n\leq N} f_n$ converge uniformly on compact subsets to a holomorphic (resp. meromorphic) function F.

Rem. 134
Zeros (resp. poles) of F lie among the zeros (resp. poles) of the f_n. If $f_n \not\equiv 0$, then $\sum_{n\geq1} f_n'/f_n$ converges normally on compact subsets to F'/F.

Def. 135 (**Euler's Constant**)

$$\gamma = \lim_{n\to\infty}\left(-\log n + \sum_{k=1}^{n}\frac{1}{k}\right).$$

Def. 136 (**Gamma Function**)

$$\Gamma(z) = z^{-1}e^{-\gamma z}\prod_{k=1}^{\infty}\left(1 + \frac{z}{k}\right)^{-1} e^{z/k}.$$

Thm. 137 (**Properties of the Gamma Function**)
Γ is meromorphic on $\mathbb{C}$, holomorphic on $\mathbb{C} \setminus \mathbb{Z}_{\leq0}$, and has no zeros.

Rem. 138 (**Poles and Residues of** Γ)
For every $n \in \mathbb{N}$, Γ has a simple pole at $-n$ with residue

$$\operatorname{Res}(\Gamma, -n) = \frac{(-1)^n}{n!}.$$

Rem. 139
$\Gamma(z+1) = z\Gamma(z)$. Consequently $\Gamma(1) = 1$, $\Gamma(n+1) = n!$.

Rem. 140 (**Relation with** ζ)
For $\Re(s) > 1$,

$$\zeta(s) = \frac{1}{\Gamma(s)}\int_0^\infty \frac{t^{s-1}}{e^t - 1}\, dt.$$

Rem. 141 (**Analytic Properties of** ζ)
ζ extends to a meromorphic function on $\mathbb{C}$ with a simple pole at $s = 1$ and residue 1.

ζ has simple zeros at $2\mathbb{Z}_{<0}$.
All other zeros lie in the strip $0 \leq \Re(s) \leq 1$.

T1

Def. 142 (**Isolated Point**)
$x \in X$ is isolated if $\{x\}$ is open.

Prop. 143
If X is connected and $f : X \to \mathbb{Q}$ is continuous, then f is constant.

Prop. 144 (**Linear Maps on Normed Spaces**)
For normed spaces V, W over $\mathbb{R}$, a linear map $L : V \to W$ is continuous iff it is bounded, i.e. $\exists A > 0$ such that $\|L(v)\| \leq A\|v\|$.

Prop. 145 (**Equivalence of Norms**)
On a finite-dimensional vector space, any two norms define the same topology.

Prop. 146
The closed unit ball of a normed vector space is compact iff the space is finite-dimensional.

Prop. 147
$C^0([0,1])$ with $\|f\|_0 = \sup_{[0,1]}|f(x)|$ is complete.

Prop. 148
Let (X, d) be metric. Limits are unique. $Y \subset X$ is closed iff it contains limits of convergent sequences in Y. $\overline{Y} = Y \cup \{\text{accumulation points of } Y\}$.

Prop. 149
$Y \subset X$ metric is disconnected iff $\exists$ open $U, V \subset X$ with $Y \subset U \cup V$, $Y \cap U \neq \varnothing$, $Y \cap V \neq \varnothing$, $U \cap V = \varnothing$.

Prop. 150
If $f_n : X \to Y$ are continuous, pointwise convergence need not preserve continuity, but local uniform convergence implies continuity of the limit.

Def. 151 (**Cauchy Sequence**)
(x_n) is Cauchy if $\forall\varepsilon > 0\ \exists N$ such that $d(x_n, x_m) < \varepsilon$ for $n, m \geq N$. X is complete if every Cauchy sequence converges.

Prop. 152
Finite products of complete metric spaces are complete.

Prop. 153
$\mathbb{R}^N$ is complete.

Prop. 154
Cauchy sequences in $\mathbb{Q}$ modulo $(x_n) \sim (y_n)$ if $x_n - y_n \to 0$ give $\mathbb{R}$.

Thm. 155 (**Baire Category Theorem**)
If (X, d) is complete and U_n are dense open sets, then $\bigcap_{n=1}^{\infty} U_n$ is dense.

HW1

Def. 156 (**Dense Set**)
$A \subset X$ is dense iff $\overline{A} = X$.

Def. 157 (**Lipschitz Map**)
$f : (X, d_X) \to (Y, d_Y)$ is Lipschitz if $d_Y(f(x), f(x')) \leq A d_X(x, x')$. Every Lipschitz map is continuous.

Thm. 158
A non-empty complete metric space without isolated points is uncountable.

HW2

Def. 159
$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$, $\sin z = \sum_{n=0}^{\infty}(-1)^n \frac{z^{2n+1}}{(2n+1)!}$,
$\cos z = \sum_{n=0}^{\infty}(-1)^n \frac{z^{2n}}{(2n)!}$, $\tan z = \frac{\sin z}{\cos z}$.

Def. 160 (Laplacian)
$\Delta = \partial_x^2 + \partial_y^2$.

Thm. 161 (Taylor's Expansion)
Let f be C^2 near z_0. Then, as $z \to z_0$,

$$\begin{aligned} f(z) &= f(z_0) + f_z(z_0)\,(z-z_0) + f_{\bar z}(z_0)\,(\bar z - \bar z_0) \\ &\quad + \frac{1}{2} f_{zz}(z_0)\,(z-z_0)^2 + f_{z\bar z}(z_0)\,(z-z_0)(\bar z - \bar z_0) \\ &\quad + \frac{1}{2} f_{\bar z \bar z}(z_0)\,(\bar z - \bar z_0)^2 + o(|z-z_0|^2). \end{aligned}$$

Def. 162 (Harmonic Function)
u is harmonic iff $\Delta u = 0$.

Def. 163 (Harmonic Conjugate)
Let u be harmonic on D. v is called a harmonic conjugate of u if $f = u + iv$ is holomorphic on D.

Thm. 164
If u is harmonic on a simply connected domain D, then $u = \Re(f)$ for some holomorphic f on D. unique up to an additive constant.

Def. 165 (Wirtinger Operators)
$\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$, $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$.

Lem. 166
$\partial_z z = 1$, $\partial_z \bar z = 0$, $\partial_{\bar z} z = 0$, $\partial_{\bar z} \bar z = 1$.

Thm. 167
f holomorphic $\iff \partial_{\bar z} f = 0$.

Def. 168 (Anti-Holomorphic)
f is anti-holomorphic $\iff \partial_z f = 0$.

Thm. 169
$\Delta = 4\,\partial_z \partial_{\bar z}$.

Thm. 170
u harmonic $\iff \partial_{\bar z} u$ is holomorphic.

Thm. 171
If f is holomorphic, then $\Re f$ and $\Im f$ are harmonic.

Thm. 172
If u is harmonic and f holomorphic, then $u \circ f$ is harmonic.

Thm. 173
If f is holomorphic and non-vanishing, then $\log|f|$ is harmonic.

Thm. 174 (Cauchy–Riemann Equations)
$f = u + iv$ is holomorphic iff $u_x = v_y$ and $u_y = -v_x$.

T4

Thm. 175 (Phragmén–Lindelöf Principle (Sector Version))
Let

$$\Omega = \{z = re^{i\theta} : r > 0,\ |\theta| < \alpha\}, \quad 0 < \alpha < \pi.$$

Let f be holomorphic in Ω and continuous on $\overline{\Omega}$. Assume:

1. $|f(z)| \le M$ for $z \in \partial\Omega$,
2. There exists $C > 0$ and $\beta < \frac{\pi}{2\alpha}$ such that

$$|f(z)| \le Ce^{|z|^\beta} \quad \text{for all } z \in \Omega.$$

Then

$$|f(z)| \le M \quad \text{for all } z \in \Omega.$$

Thm. 176 (Schwarz Reflection Principle)
Let $U \subset \mathbb{C}$ be a domain symmetric about $\mathbb{R}$. Let

$$U^+ = \{z \in U : \Im z > 0\}.$$

Suppose f is holomorphic on U^+, continuous on $U^+ \cup (U \cap \mathbb{R})$, and satisfies

$$f(x) \in \mathbb{R} \quad \text{for } x \in U \cap \mathbb{R}.$$

Define

$$F(z) = \begin{cases} f(z), & \Im z \ge 0, \\ \overline{f(\overline{z})}, & \Im z < 0. \end{cases}$$

Then F is holomorphic on U.

T5

Thm. 177 (Schwarz Lemma)
Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Then

$$|f(z)| \le |z| \quad \text{for all } z \in \mathbb{D},$$

and

$$|f'(0)| \le 1.$$

Moreover, if equality holds for some $z \neq 0$ or if $|f'(0)| = 1$, then $f(z) = e^{i\theta}z$ for some $\theta \in \mathbb{R}$.

Thm. 178 (Disk Automorphism)
For $a \in \mathbb{D}$, define

$$\varphi_a(z) = \frac{a - z}{1 - \overline{a}z}.$$

Then $\varphi_a : \mathbb{D} \to \mathbb{D}$ is a biholomorphic map with

$$\varphi_a(a) = 0, \quad \varphi_a(0) = a, \quad \varphi_a^{-1} = \varphi_a.$$

Rem. 179
Every biholomorphic automorphism of $\mathbb{D}$ is of the form

$$f(z) = e^{i\theta}\,\varphi_a(z), \quad a \in \mathbb{D},\ \theta \in \mathbb{R}.$$

T6

Thm. 180 (Cauchy–Hadamard Formula)
Let $\sum_{n\ge0} a_n z^n$ be a power series. The radius of convergence R is given by

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

Rem. 181 (Standard Möbius Transformations)

$$\varphi(z) = \frac{1+z}{1-z} \quad : \quad \mathbb{D} \to \{w : \Re w > 0\}, \qquad \varphi(0) = 1.$$

$$\psi(w) = \frac{w-1}{w+1} \quad : \quad \{w : \Re w > 0\} \to \mathbb{D}, \qquad \psi(1) = 0.$$

Thm. 182 (Riemann Mapping Theorem)
Let $\Omega \subsetneq \mathbb{C}$ be a simply connected domain with $\Omega \neq \mathbb{C}$. Then there exists a biholomorphic map $f : \Omega \to \mathbb{D}$.

T7

Thm. 183 (Mean Value Property and Maximum Principle)
Let D be a domain and $u : D \to \mathbb{R}$ harmonic.
If $\overline{B_r(a)} \subset D$, then

$$u(a) = \frac{1}{2\pi}\int_0^{2\pi} u(a + re^{i\theta})\,d\theta.$$

Hence u satisfies the maximum principle: if u attains a local maximum (or minimum) in D, then u is constant.

Thm. 184 (Schwarz Integral Formula)
Let f be holomorphic in a neighbourhood of $\overline{\mathbb{D}}$. Then for $z \in \mathbb{D}$,

$$f(z) = i\,\Im f(0) + \frac{1}{2\pi}\int_0^{2\pi} \frac{e^{it}+z}{e^{it}-z}\,\Re f(e^{it})\,dt.$$

Rem. 185 (Identify Infinite Sums/Products)
Identify infinite sums/products by: normal convergence on compact sets; locate poles/zeros and orders; check parity/periodicity; compare principal parts; subtract/divide by candidate; get an entire bounded function; apply Liouville; determine constant at a convenient point.

HW4

Def. 186 (Conformal Map)
Let $\Omega, \Omega' \subset \mathbb{C}$ be domains. $f : \Omega \to \Omega'$ is conformal if it is holomorphic and bijective. Then f^{-1} is also holomorphic.

Def. 187 (Automorphism Group)

$$\mathrm{Aut}(\Omega) := \{f : \Omega \to \Omega \text{ biholomorphic}\}.$$

Thm. 188 (Riemann Uniformization Theorem)
Let $\Omega \subsetneq \mathbb{C}$ be simply connected and $a \in \Omega$. Then $\exists$ a unique conformal map $\phi : \mathbb{D} \to \Omega$ such that $\phi(0) = a$, $\phi'(0) > 0$. Also, $\phi^{-1} : \Omega \to \mathbb{D}$ is holomorphic.

Def. 189 (Blaschke Product)
Let $(a_n) \subset \mathbb{D}$ with $\sum_{n\ge1}(1 - |a_n|) < \infty$. Define

$$f_n(z) = \begin{cases} \dfrac{|a_n|}{a_n}\,\dfrac{a_n - z}{1 - \overline{a_n}z}, & a_n \neq 0, \\ z, & a_n = 0, \end{cases} \qquad F(z) = \prod_{n\ge1} f_n(z).$$

Thm. 190
The infinite product F converges normally on compact subsets of $\mathbb{D}$, hence defines a holomorphic function on $\mathbb{D}$. $|F(z)| < 1$ for all $z \in \mathbb{D}$. The zeros of F are exactly the points a_n, with multiplicity. Each factor satisfies $|f_n(z)| = 1$ for $|z| = 1$, so F is bounded by 1 in $\mathbb{D}$.

T8

Thm. 191
A positive-dimensional complex submanifold of $\mathbb{C}^n$ is not compact.

Rem. 192 (Generalized Argument Principle)
Let f be holomorphic in a region containing $\overline{D}$, and assume f has no zeros on ∂D. If $a_1, \ldots, a_m$ are the zeros of f in D, counted with multiplicity, then for every holomorphic function ϕ on a neighbourhood of $\overline{D}$,

$$\frac{1}{2\pi i}\int_{\partial D} \phi(z)\frac{f'(z)}{f(z)}\,dz = \sum_{j=1}^{m}\phi(a_j).$$

Thm. 193 (Liouville for Harmonic Functions)
A bounded harmonic function on $\mathbb{C}$ is constant.

Thm. 194
Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(a) = f(-a) = 0$. Then

$$|f(0)| \le |a|^2.$$

If equality holds, then

$$f(z) = e^{i\theta}\,\varphi_a(z)\varphi_{-a}(z)$$

for some $\theta \in \mathbb{R}$.

Lem. 195
Let f, g be entire functions. If

$$|f(z)| \le |g(z)| \qquad \forall z \in \mathbb{C},$$

then there exists $c \in \mathbb{C}$ with $|c| \le 1$ such that

$$f = cg.$$

Thm. 196 (Automorphisms of $\mathbb{C}$)
Every biholomorphic map $f : \mathbb{C} \to \mathbb{C}$ is of the form

$$f(z) = az + b, \qquad a \neq 0.$$

Thm. 197 (Automorphisms of $\mathbb{C} \setminus \{0,1\}$)
Every biholomorphic map

$$f : \mathbb{C} \setminus \{0,1\} \to \mathbb{C} \setminus \{0,1\}$$

permutes $\{0, 1, \infty\}$. Hence

$$\mathrm{Aut}(\mathbb{C} \setminus \{0,1\}) \cong S_3,$$

and the automorphisms are exactly

$$z, \quad 1-z, \quad \frac{1}{z}, \quad \frac{z-1}{z}, \quad \frac{1}{1-z}, \quad \frac{z}{z-1}.$$

Thm. 198 (Automorphisms of $\mathbb{P}^1$)
Every holomorphic automorphism of $\mathbb{P}^1$ is a Möbius transformation:

$$f(z) = \frac{az+b}{cz+d}, \qquad ad - bc \neq 0.$$

Hence

$$\mathrm{Aut}(\mathbb{P}^1) \cong \mathrm{PGL}_2(\mathbb{C}).$$

T9

Thm. 199
Let Ω be a bounded domain. If f is holomorphic and never zero in Ω, continuous on $\overline{\Omega}$, and $|f|$ is constant on $\partial\Omega$, then f is constant.

Rem. 200
A standard conformal map

$$\mathbb{C} \setminus \big((-\infty, a] \cup [b, \infty)\big) \to \mathbb{C} \setminus [0, \infty)$$

is obtained by

$$z \mapsto \frac{z-a}{z-b},$$

with a suitable branch of the square root.

Def. 201 (Poisson Kernel)

$$P(z, e^{it}) = \frac{1 - |z|^2}{2\pi|z - e^{it}|^2}.$$

Thm. 202 (Poisson Integral Formula on $\mathbb{D}$)
If u is harmonic in a neighbourhood of $\overline{\mathbb{D}}$, then for $z \in \mathbb{D}$,

$$u(z) = \int_0^{2\pi} \frac{1 - |z|^2}{2\pi|z - e^{it}|^2}\,u(e^{it})\,dt.$$

The same formula holds if u is continuous on $\overline{\mathbb{D}}$ and harmonic on $\mathbb{D}$.

Thm. 203 (Harnack Inequality on $\mathbb{D}$)
If u is harmonic in a neighbourhood of $\overline{\mathbb{D}}$ and $u \ge 0$ on $\partial\mathbb{D}$, then for $|z| < 1$,

$$\frac{1 - |z|}{1 + |z|}\,u(0) \le u(z) \le \frac{1 + |z|}{1 - |z|}\,u(0).$$

Thm. 204 (Dirichlet Problem on $\mathbb{H}$)
Let $\mathbb{H} = \{z \in \mathbb{C} : \Im z > 0\}$. If $u : \mathbb{H} \cup \mathbb{R} \to \mathbb{R}$ is continuous, harmonic on $\mathbb{H}$, and has a finite limit at ∞, then for $z_0 \in \mathbb{H}$,

$$u(z_0) = \Re\left(\frac{1}{\pi i}\int_{-\infty}^{\infty} \frac{u(t)}{t - z_0}\,dt\right).$$

Equivalently,

$$u(x+iy) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{y\,u(t)}{(t-x)^2 + y^2}\,dt.$$